

Question

What are you trying to discover?

Which state had the most wildfires in 2017?

Description of Data Set

https://www.predictiveservices.nifc.gov/intelligence/2017_statssumm/intro_summary17.pdf

The data set I chose lists significant wildland fires over 40,000 acres in 2017.

Hypothesis

My hypothesis is that states near the west coast will have more wildfires like California or Nevada.

Methods

```
In [9]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
wf = pd.read_csv('wf_2017.csv')
print(wf.shape)
```

(51, 8)

```
In [10]: fires = wf['State']  
fires
```

```
Out[10]: 0      OK  
        1      TX  
        2      CA  
        3      MT  
        4      NV  
        5      OR  
        6      MT  
        7      NV  
        8      GA  
        9      TX  
       10      WA  
       11      CA  
       12      MT  
       13      NV  
       14      NV  
       15      AK  
       16      CA  
       17      ID  
       18      CA  
       19      CA  
       20      CA  
       21      AK  
       22      CA  
       23      MT  
       24      UT  
       25      OR  
       26      CA  
       27      NV  
       28      MT  
       29      NV  
       30      TX  
       31      ID  
       32      AK  
       33      SD  
       34      MT  
       35      OR  
       36      WA  
       37      CA  
       38      ID  
       39      CA  
       40      OR  
       41      AZ  
       42      AZ  
       43      AK  
       44      MT  
       45      AK  
       46      ID  
       47      MT  
       48      NV  
       49      AK  
       50      AK  
Name: State, dtype: object
```

```
In [11]: print(wf)
```

	NAME	GACC	State	Start Date	Contain or Last report date \
0	NW Oklahoma Complex	SA	OK	3/7	3/24
1	Perryton	SA	TX	3/6	3/13
2	Thomas	SO	CA	12/4	Active into 2018
3	Lodgepole Complex	NR	MT	7/20	8/11
4	Roosters Comb	GB	NV	7/9	7/24
5	Chetco Bar	NW	OR	7/12	10/26
6	Rice Ridge	NR	MT	7/24	10/17
7	Four Seasons Complex	GB	NV	7/16	7/25
8	West Mims	SA	GA	4/6	6/29
9	Lefors East	SA	TX	3/7	3/10
10	Diamond Creek	NW	WA	7/23	10/5
11	Central LNU Complex	NO	CA	10/9	10/10
12	Sartin Draw	NR	MT	8/30	9/4
13	Truckee	GB	NV	7/3	7/11
14	Tohakum 2	GB	NV	8/29	7/11
15	Campbell River	AK	AK	6/26	10/12
16	Oak	NO	CA	8/12	9/23
17	Highline	GB	ID	7/28	9/21
18	Long Valley	GB	CA	7/11	7/21
19	Modoc July Complex	NO	CA	7/24	8/13
20	Detwiler	SO	CA	7/16	8/7
21	White Mountain Creek	AK	AK	7/3	10/5
22	Eclipse Complex	NO	CA	8/15	10/8
23	Brush Flat	NR	MT	8/31	9/4
24	Brianhead	GB	UT	6/17	7/24
25	Nena Springs	NW	OR	8/8	8/29
26	Salmon August Complex	NO	CA	8/13	10/4
27	Dry Gulch	GB	NV	7/4	7/11
28	Meyers	NR	MT	7/14	10/13
29	Snowstorm	GB	NV	7/14	7/15
30	Rankin Ranch Rd.	SA	TX	3/23	3/25
31	Powerline	GB	ID	8/4	8/13
32	Nowitna	AK	AK	7/14	8/31
33	Legion Lake	RM	SD	12/11	12/19
34	Lolo Peak	NR	MT	7/15	10/6
35	Cinder Butte	NW	OR	8/2	8/16
36	Norse Peak	NW	WA	8/11	10/5
37	Southern LNU Complex	NO	CA	10/9	10/27
38	Mammoth Cave	GB	ID	8/4	8/8
39	Garza	SO	CA	7/9	7/10
40	Eagle Creek	NW	OR	9/2	10/27
41	Frye	SW	AZ	6/7	8/31
42	Sawmill	SW	AZ	4/23	5/3
43	Bear Mountain	AK	AK	7/9	11/16
44	Sapphire Complex	NR	MT	7/24	10/4
45	Umpqua North Complex	NW	AK	8/11	10/12
46	Powerline	NR	ID	7/15	7/29
47	Highway 200 Complex	NR	MT	9/1	10/6
48	Earthstone	GB	NV	7/3	7/11
49	Boulder Creek	AK	AK	7/2	10/12
50	Helmet	AK	AK	7/6	8/24

Size (Acres) Cause Estimated Cost

0 779,292 U 3,200,000

1	318,156	H	NR
2	270,000	U	123,836,000
3	270,000	U	9,800,000
4	218,380	U	4,000,000
5	191,125	L	72,000,000
6	160,187	L	49,251,000
7	159,986	U	4,573,000
8	152,515	L	45,500,000
9	135,000	H	NR
10	128,272	U	14,760,000
11	110,720	U	102,000,000
12	99,735	L	1,500,000
13	98,960	U	2,120,260
14	94,221	L	3,456,000
15	93,520	L	1,056,688
16	86,830	L	NR
17	84,619	L	2,167,864
18	83,733	U	11,953,000
19	83,120	U	35,200,000
20	81,826	U	90,000,000
21	78,857	L	96,310
22	78,698	U	46,006,691
23	73,797	L	400,000
24	71,673	H	3,660,000
25	68,007	H	8,900,000
26	65,889	U	39,805,700
27	64,000	U	2,700,000
28	62,034	L	32,800,000
29	60,000	U	482,448
30	60,000	H	NR
31	55,529	U	4,512,207
32	55,254	L	307,802
33	54,023	H	2,200,000
34	53,902	L	48,500,000
35	52,531	H	4,400,000
36	52,056	U	19,760,000
37	51,624	U	48,509,895
38	49,912	H	560,000
39	48,889	U	15,453,433
40	48,831	H	20,251,700
41	48,443	L	20,000,000
42	46,991	H	8,200,000
43	46,715	L	NR
44	43,733	U	35,406,736
45	43,158	U	42,890,561
46	42,558	L	2,958,255
47	42,442	L	12,500,000
48	41,545	U	3,194,000
49	40,906	L	917,520
50	40,202	L	26,978

```
In [12]: states = wf.groupby("State")["NAME"].count()  
states
```

```
Out[12]: State  
AK      7  
AZ      2  
CA     10  
GA      1  
ID      4  
MT      8  
NV      7  
OK      1  
OR      4  
SD      1  
TX      3  
UT      1  
WA      2  
Name: NAME, dtype: int64
```

```
In [13]: states = wf.groupby("State")["State"].count().max()  
states
```

```
Out[13]: 10
```

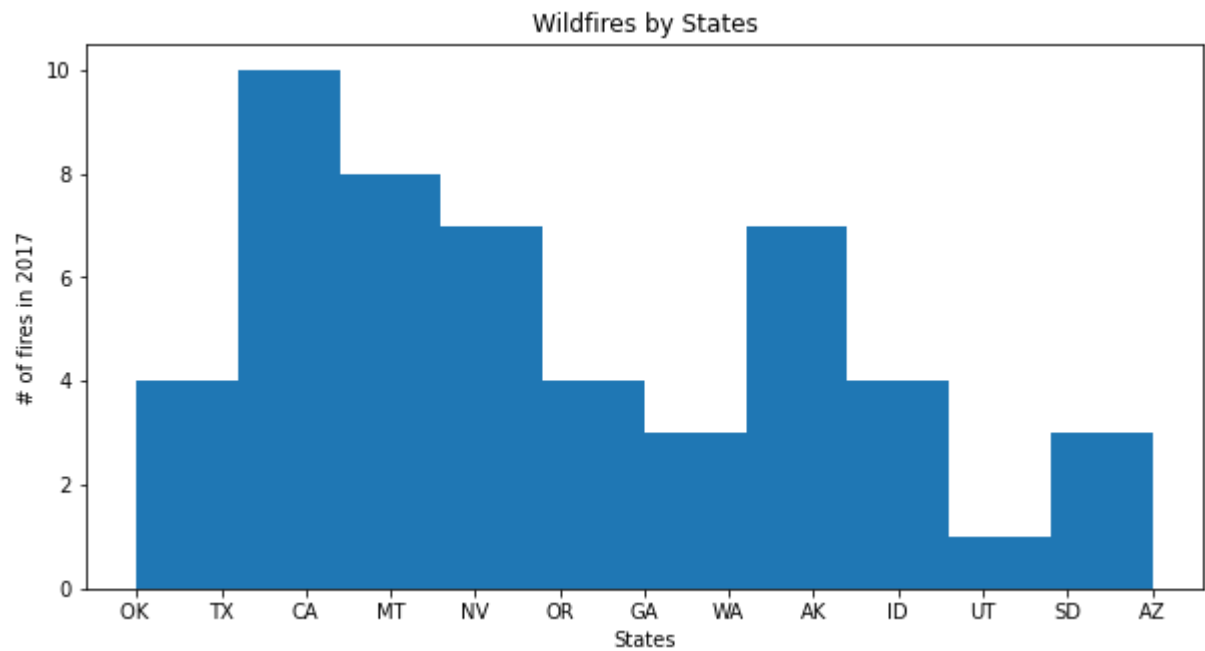
```
In [14]: states = wf.groupby("State")["State"].count().idxmax()  
states
```

```
Out[14]: 'CA'
```

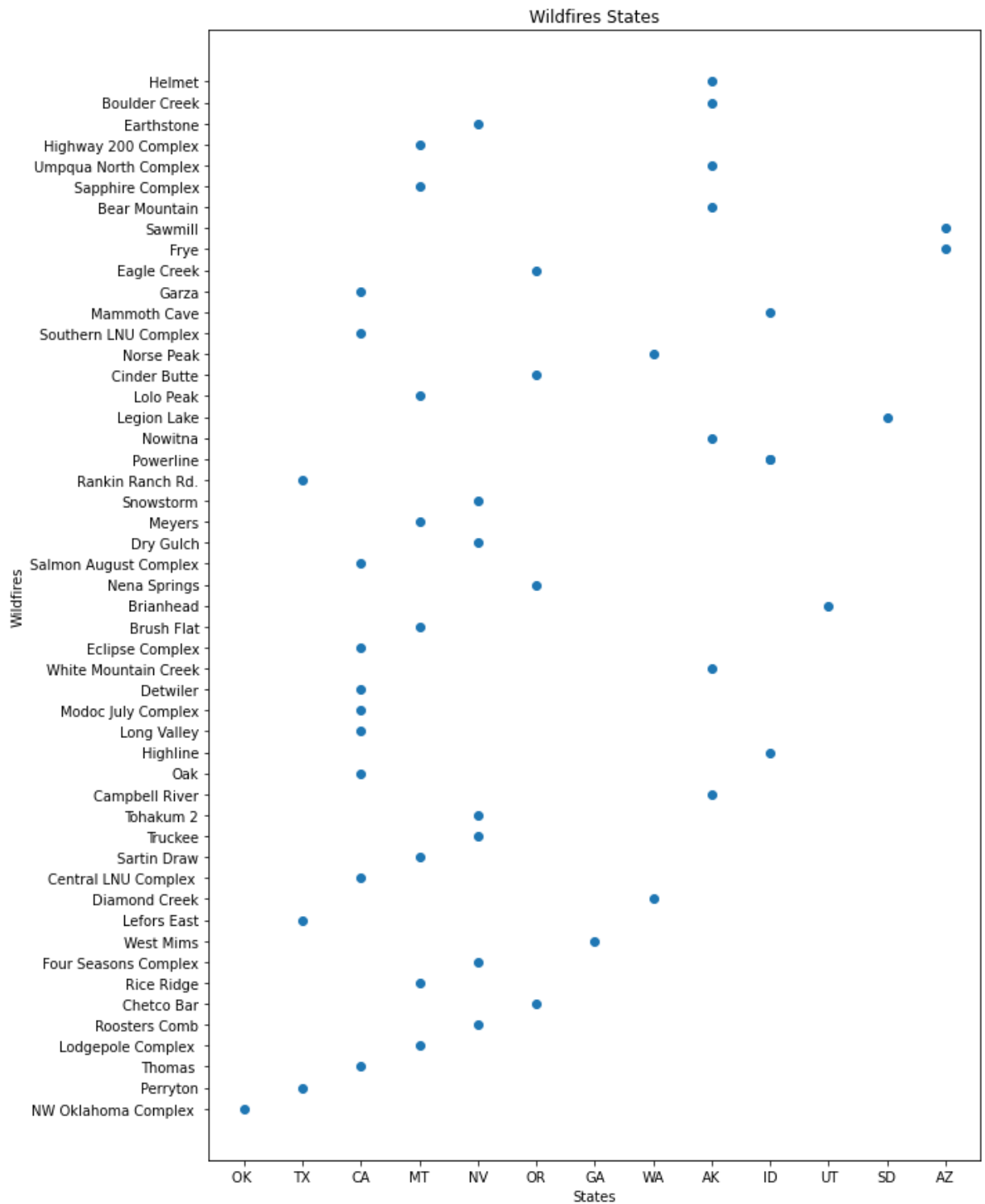
Visualizations

```
In [15]: plt.figure(figsize=(10,5))
plt.hist(wf['State'])
plt.xlabel('States')
plt.ylabel('# of fires in 2017')
plt.title('Wildfires by States')

plt.show()
```



```
In [16]: plt.figure(figsize=(10,15))
plt.scatter(wf['State'], wf['NAME'])
plt.xlabel('States')
plt.ylabel('Wildfires')
plt.title('Wildfires States')
plt.show()
```



Conclusions

I can conclude that my hypothesis was correct, California resulted with 10 being the state with the most wildfire records. California had the most recorded data in America.

Further Research

I want to look into places where climate change affected the weather and natural disasters like wild fires to occur.